

CIS 635

Knowledge Discovery & Data Mining

Fall 2026 Syllabus

Updated: September 1, 2026

Instructor: Dr. Kamrul Hasan

Credits: 3

Pronouns: he/him/his

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Meeting Times: Section 01: T 6:00 pm–8:50 pm; Section 02 MW 4:30 pm–5:45 pm (DCIH 420)

Office Hours: Online / MAK D-2-216
([schedule by appointment](#))

1. General Information

1.1 Course Objectives

Upon completion of this course, students will be able to:

- Assess key models, methods, and concepts for data mining and knowledge discovery across domains such as healthcare, life sciences, business, and other applied areas.
- Evaluate data mining software and tools for capability, validity, reproducibility, and responsible use.
- Apply statistical and algorithmic methods, including database search programs and advanced techniques, to analyze diverse datasets.
- Formulate database queries for common relational and nonrelational databases across health, biomedical, business, and other applied domains.
- Wrangle, analyze, and visualize data, primarily using Python, and communicate findings using effective visualizations.
- Identify, build, and evaluate data mining models for their effectiveness and appropriateness, and communicate defensible conclusions based on data, model performance, assumptions, and limitations.

1.2 Course Description

This course teaches students to extract knowledge from data by solving real-world analytical problems, emphasizing the full data mining process, from data preparation to actionable insight, and developing sound judgment in applying data mining and knowledge discovery methods across domains such as healthcare, life sciences, business, and other applied areas. Particular emphasis is placed on applications in bioinformatics and health-related data: students gain hands-on experience with biological and biomedical databases, dynamic programming for sequence alignment and similarity search (e.g., BLAST), hidden Markov models for sequence and structure analysis, and clustering and classification methods for gene-expression data. Students also build practical

skills in Python-based data analysis, data preparation and visualization, predictive modeling, model evaluation, clustering, dimensionality reduction, association analysis, and neural networks, and gain experience formulating queries against relational and nonrelational databases while completing end-to-end knowledge discovery tasks involving real-world datasets.

1.3 Requirements and Prerequisite

Students need an internet connection, a computer, access to Blackboard, and the ability to run Python and install required Python packages. If you have difficulty accessing these resources or meeting these requirements, contact the instructor so that an appropriate solution can be explored.

Prerequisite: N/A

1.4 Delivery Format

The course is delivered primarily through in-person class meetings, supported by Blackboard for course materials, assignments, announcements, and grades.

1.5 Recommended References

Specific references will be shared throughout the course. The following resources are recommended:

- *Data Mining: Concepts and Techniques*, 4th Edition, by Jiawei Han, Jian Pei, and Hanghang Tong, Morgan Kaufmann, 2023.
- *Data Mining: A Knowledge Discovery Approach* by Witold Pedrycz and collaborators (available online through the GVSU Library).
- *Introduction to Data Mining*, 2nd Edition, by Pang-Ning Tan, Michael Steinbach, and Vipin Kumar.
- *Pattern Recognition and Machine Learning* by Christopher M. Bishop.
- *Fundamentals of Data Visualization* by Claus O. Wilke.
- *Python Data Visualization Essentials Guide* by Kalilur Rahman.
- Selected papers from leading data mining and knowledge discovery venues (e.g., KDD, ICDM, CIKM, EMNLP) will be assigned throughout the semester.

2. Tentative Schedule

Subject to change.

Week	Week of	Topics
1	Aug. 31	Foundations of Knowledge Discovery & the Data Mining Process; Python Basics
2	Sept. 7	Core Concepts in Data, Statistics, & Computational Thinking; Descriptive Statistics & Data Visualization
3	Sept. 14	Core Concepts in Data, Statistics, & Computational Thinking (cont'd); Exploratory Data Analysis (EDA) I

Week	Week of	Topics
4	Sept. 21	Data Preparation, Similarity & Distance Measures, & Feature Selection; EDA II & Case Studies
5	Sept. 28	Dimensionality Reduction; K-Nearest Neighbors
6	Oct. 5	Data Mining Approaches: Predictive Modeling – Regression
7	Oct. 12	Data Mining Approaches: Predictive Modeling – Classification I
8	Oct. 19	Data Mining Approaches: Predictive Modeling – Classification II & Model Evaluation
9	Oct. 26	<i>Fall Break</i>
10	Nov. 2	Data Mining Approaches: Clustering
11	Nov. 9	Data Mining Approaches: Association Analysis & Anomaly Detection
12	Nov. 16	Working with Databases & Large Datasets: Relational & Nonrelational Query Formulation, incl. Biomedical & Health Databases
13	Nov. 23	Evaluation, Validation, & Interpretation of Results; Sequence Analysis & Modeling (Bioinformatics: BLAST, Hidden Markov Models)
14	Nov. 30	Contemporary Topics & Trends in Data Mining: Deep Learning & Neural Networks
15	Dec. 7	Applied Knowledge Discovery Across Domains (Health, Bioinformatics, Business); Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week

Final Examination Period

The CIS 635 final examination will be administered during the University examination period. Class instruction ends December 12, 2026, and the examination period is December 12–19, 2026. The official final-exam date, time, and location will follow the University final-examination schedule and will be communicated through Blackboard.

3. Course Organization and Grading Format

3.1 Grading Distribution

The final course grade will be determined using the following grading distribution:

Assessment	Weight
Exams	40%
Team KDD project (incl. final presentation)	30%
Labs, programming assignments, & quizzes	30%
Total	100%

3.1.1 Homework Assignments and Tests

Labs, programming assignments, and quizzes will reinforce course material, and exams will typically include a mixture of programming problems and mathematical or written questions.

Unless otherwise stated, individual assignments must be completed independently. Students may discuss higher-level concepts and ask questions, but may not share code or responses or review another student's code or responses.

3.1.2 Final Project

Students will work in groups on a knowledge discovery and data mining project. Each group will identify a problem and associated dataset(s) related to a domain of expertise and/or interest (e.g., health, bioinformatics, or business) and complete an end-to-end KDD project. The project emphasizes a learning-by-doing approach.

Each team will deliver a final presentation and submit project documentation; the presentation is graded as part of the Team KDD project weight above. All members must explicitly describe their individual and collective contributions to the project.

3.2 Grading Scale

For final grade calculation, any fractional final percentage will be rounded up to the next whole percentage. For example, 89.1% will be recorded as 90%.

Grade	Minimum	Grade	Minimum
A	93%	C	73%
A-	90%	C-	70%
B+	87%	D+	67%
B	83%	D	60%
B-	80%	F	Below 60%
C+	77%		

4. Course Expectations, Policies, and Academic Honesty

4.1 Expectations

To support your success in this course, you are expected to:

- Check Blackboard regularly for announcements, course materials, assignments, and grades.
- Stay up to date with required course videos and readings.
- Communicate feedback about the course and teaching that may help improve the learning experience.
- Adhere to CIS and GVSU policies on academic honesty.

4.2 Course Policies

- Weekly reflections and homework assignments must be completed individually unless otherwise stated.
- **Due dates:** Specific due dates will be posted with each assignment in Blackboard. Unless otherwise announced, assignments are due at 11:59 p.m. Eastern Time on the stated due date.

- **Late policy:** 10% of the maximum possible grade is deducted per day late, up to three days (30% total). After three days, the assignment will not be accepted.
- The instructor reserves the right to modify course policies, the course calendar, and due dates as needed.
- This course is subject to the GVSU course policies listed at www.gvsu.edu/coursepolicies/.

4.3 Use of Generative AI Tools

Generative AI tools may be used only as permitted for individual course activities and assignments. Students remain responsible for understanding, validating, and being able to explain all submitted work, including code, analyses, visualizations, and written material. AI-generated content or code must not be submitted as the student's own work when prohibited by assignment instructions. When AI assistance is used in a permitted manner, students may be required to disclose the tool used and the nature of the assistance. Specific AI-use expectations will be provided with individual assignments and assessments.

4.4 Academic Honesty

All students are expected to adhere to the academic honesty standards of Grand Valley State University and the academic honesty guidelines of the School of Computing. Details are available at <https://www.gvsu.edu/computing/academic-honesty-30.htm>.

Collaboration and learning from peers and the broader community are encouraged, but presenting another person's work as your own is prohibited. For individual assignments:

- Document collaboration.
- Do not electronically transfer code between students.
- Cite code found on the internet with an active link to the source. External code must not solve the entirety of an assigned problem or project.
- You may participate in online forums, but do not post solutions or solicit others to complete your work.
- You may discuss problems with peers in non-technical terms, but you remain responsible for your own submission.
- You are ultimately responsible for all aspects of your submissions and should be able to explain and defend your work.

5. Useful Resources

5.1 Blackboard

Course materials, assignments, grades, and announcements will be posted to Blackboard (<https://lms.gvsu.edu/>). Students are responsible for staying informed. Blackboard announcements are used to communicate timely information, including changes to deadlines; students should ensure their notification settings allow them to receive updates.

5.2 Other Academic Resources

GVSU provides academic-support resources, including:

- The Writing Center
- Speech Lab
- Research Consultants
- Library Liaisons

5.3 Disability Support

Students who need accommodations should present a memo from Disability Support Resources (DSR) describing the approved reasonable accommodations. If you have not already done so, contact Disability Support Resources at 215 CON, by phone at 331-2490, or by email at dsrgvsu@gvsu.edu. The instructor cannot provide disability accommodations until a DSR-issued memo has been received. All discussions will remain confidential. More information is available at <https://www.gvsu.edu/dsr/>.

5.4 Religious Observance

Many University students, staff, and faculty observe religious traditions. The Religious Inclusion Policy acknowledges the right of students, staff, and faculty to engage in religious observances. If you need special accommodations to observe a religious holiday, please make arrangements with the instructor in advance.

5.5 In Case of Emergency

Fire: Immediately proceed to the nearest exit when a fire alarm sounds. Do not use elevators. Additional information is available on the University Emergency website at gvsu.edu/emergency.